usepackage[a4paper,margin=0.65in]geometry

[766] (CONTINUED)

ON THE GEODESIC CURVATURE OF A CURVE ON A SURFACE.

The Special Curves, p constant and q constant.

Consider the curve $p=$ const. For this curve $p'=0$, $p''=0$; therefore also $Gq'^2=1$, and, if R be the radius of geodesic curvature, then

$$\frac{1}{R}=\tfrac{1}{V}\,\mathfrak{D}q'^3,\qquad =\tfrac{1}{V}\,\frac{\mathfrak{D}}{G\sqrt{G}}.$$

Similarly for the curve $q=$ const. Here $q'=0$, $q''=0$; therefore $Ep'^2=1$, and, if S be the radius of geodesic curvature, then

$$\frac{1}{S}=\tfrac{1}{V}\,\mathfrak{A}p'^3,\qquad =\tfrac{1}{V}\,\frac{\mathfrak{A}}{E\sqrt{E}}.$$

These values of R and S are interesting for their own sakes, and they will be introduced into the expression for the radius of geodesic curvature ρ of the general curve.

Transformed Formula for the Radius of Geodesic Curvature.

From the values of $\frac{1}{R}$, $\frac{1}{S}$, we have

$$\frac{1}{\rho}=\frac{q'\sqrt{G}}{R}-\frac{p'\sqrt{E}}{S}=V(p'q''-p''q')+\tfrac{1}{V}\left\{\Omega-\tfrac{\mathfrak{A}}{E}p'^3-\tfrac{\mathfrak{D}}{G}q'^3\right\},$$

where the term in { } is

$$\Omega=\mathfrak{B}p'^3-\tfrac{\mathfrak{A}}{E}p'^3+\mathfrak{B}p''q'+(\mathfrak{C}p'q''+\mathfrak{D})q'^2-\tfrac{\mathfrak{D}}{G}q'^3.$$

The terms in $\mathfrak{A}$ are

$$-\tfrac{\mathfrak{A}}{E}p'(1-Ep'^2),\quad =-\tfrac{\mathfrak{A}}{E}p'(2Fp'q'+Gq'^2),$$

and those in $\mathfrak{D}$ are

$$-\tfrac{\mathfrak{D}}{G}q'(1-Gq'^2),\quad =-\tfrac{\mathfrak{D}}{G}q'(Ep'^2+2Fp'q').$$

Hence the whole expression contains the factor $p'q'$, and is, in fact,

$$p'q'\left(\mathfrak{B}-\tfrac{2\mathfrak{A}F}{E}-\tfrac{\mathfrak{D}E}{G}\right)p'+p'q'\left(\mathfrak{C}-\tfrac{\mathfrak{A}G}{E}-\tfrac{2\mathfrak{D}F}{G}\right)q',$$

or substituting for $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ their values, this is

$$=p'q'\Big\{p'\left(-GE_2+EG_2+\tfrac{2F^2G_1}{E}-2FF_2-FE_2+2EF_2-\tfrac{EFG_1}{G}\right)$$
$$+q'\left(-GE_2+EG_1-\tfrac{2F^2G_2}{G}+2FF_1+FG_1-2GF_1+\tfrac{GFE_2}{E}\right)\Big\};$$

say this is

$$=p'q'(Lp'+Mq');$$

and the formula thus is

$$\frac{1}{\rho}=\frac{q'\sqrt{G}}{R}-\frac{p'\sqrt{E}}{S}=V(p'q''-p''q')+\tfrac{1}{V}p'q'(Lp'+Mq').$$

Taking ϕ, θ to be the inclination of the curve to the curves $q=$ const., $p=$ const., respectively, and $\omega\,(=\phi+\theta)$ the inclination of these two curves to each other, then

$$\cos\phi=\frac{Fp'+Gq'}{\sqrt{G}},\quad \cos\theta=\frac{Ep'+Fq'}{\sqrt{E}},\quad \cos\omega=\frac{F}{\sqrt{EG}},$$

The formula becomes in this case much more simple. We have

$$1 = Ep'^2 + Gq'^2, \quad V = \sqrt{EG}, \quad \omega = 90^\circ, \quad \sin\theta = \cos\phi;$$

and the term $Lp' + Mq'$ becomes $= \dot{E}G - E\dot{G}$, if, as before, $\dot{E}$, $\dot{G}$ denote the complete differential coefficients $E_1p' + E_2q'$ and $G_1p' + G_2q'$. The formula then is

$$\frac{1}{\rho} = \frac{\cos\phi}{R} - \frac{\sin\phi}{S} = V(p'q'' - p''q') + \tfrac{1}{V}p'q'(\dot{E}G - E\dot{G}),$$

where the values $\dfrac{1}{R}$ and $\dfrac{1}{S}$ are now $= \dfrac{\dot{G}}{G\sqrt{G}}$ and $\dfrac{\dot{E}}{E\sqrt{E}}$ respectively. But we have moreover $\phi = \tan^{-1}\dfrac{p'\sqrt{E}}{q'\sqrt{G}}$, and thence

$$\phi' = -V(p'q'' - p''q') - \tfrac{1}{2}Vp'q'(\dot{E}G - E\dot{G});$$

or the formula finally is

$$\frac{1}{\rho} - \frac{\cos\phi}{R} + \frac{\sin\phi}{S} + \phi' = 0,$$

which is Liouville's formula referred to at the beginning of the present paper. It will be recollected that ϕ' is the differential coefficient $\dfrac{d\phi}{ds}$ with respect to the arc s of the curve.

Addition.—Since the foregoing paper was written, I have succeeded in obtaining a like interpretation of the term

$$V(p'q'' - p''q') + \tfrac{1}{V}p'q'(Lp' + Mq'),$$

which belongs to the general case. I find that these terms are, in fact, $= -\phi + \omega_1 p'$; or, what is the same thing (since $\omega = \phi + \theta$ and therefore $\omega_1 p = \phi + \theta$), are $= \theta - \omega_2 q'$. It will be recollected that ϕ is the inclination of the curve to the curve $q = c$, which passes through a given point of the curve, ϕ is the variation of ϕ corresponding to the passage to the consecutive point of the curve, viz. $\phi + \phi\, ds$ is the inclination at this consecutive point to the curve $q = c + dc$, which passes through the consecutive point; ω is the inclination to each other of the curves $p = b$, $q = c$, which pass through the given point of the curve, ω_1 the variation corresponding to the passage along the curve $q = c$, viz. $\omega_1\, ds$ is the inclination to each other of the curves $p = b + db$, $q = c$; and the like as regards θ and ω_2.

For the demonstration, we have, as above,

$$\phi = \tan^{-1}\frac{Vp'}{Fp' + Gq'}, \qquad \omega = \tan^{-1}\frac{V}{F},$$

where

$$V = \sqrt{EG - F^2};$$

and moreover $Ep'^2 + 2Fp'q' + Gq'^2 = 1$. In virtue of this last equation,

$$V^2p'^2 + (Fp' + Gq')^2 = G;$$

and we have

$$\phi = -V(p'q'' - p''q') + \tfrac{1}{G}\Box,$$

where

$$\Box = (Fp' + Gq')\,p'\dot{V} - Vp'(Fp' + Gq')\cdot;$$

or, since $V^2 = EG - F^2$, and thence $2V\dot{V} = \dot{G}E - 2F\dot{F} + E\dot{G}$, we have

$$\Box = \tfrac{1}{2}p'\{(Fp' + Gq')(\dot{G}E - 2F\dot{F} + E\dot{G}) - 2(EG - F^2)(\dot{F}p' + \dot{G}q')\}.$$

Substituting herein for $\dot{E}$, $\dot{F}$, $\dot{G}$ their values $E_1p' + E_2q'$, $F_1p' + F_2q'$, $G_1p' + G_2q'$, the term in { } becomes

$$= Ip'^2 + Jp'q' + Kq'^2,$$

where

$$\begin{aligned}
I &= FGE_1 - 2EGF_1 + EFG_1,\\
J &= G^2E_2 - 2FGF_2 + (-EG + 2F^2)G_1 + FGE_1 - 2EGF_2 + EFG_2,\\
K &= G^2E_2 - 2FGF_2 + (-EG + 2F^2)G_2.
\end{aligned}$$

But from the equation $\omega = \tan^{-1}\dfrac{V}{F}$, differentiating in regard to p, we obtain

$$\omega_1 = \tfrac{1}{EGV}(FG\dot{E} - 2EG\dot{F} + EF\dot{G}) = \tfrac{1}{EGV}I;$$

or, for p writing

$$p^{-1}\{(Ep'^2+2Fp'q'+Gq'^2)V\omega_1, = Ep'(p'^2I+\tfrac{2}{p'q'}p'^2q'^2J+\tfrac{1}{p'^2q'^2}q'^2K)\},$$

we have

$$\phi-\omega_1p' = -V(p'q''-p''q')+\tfrac{1}{G}(Ip'^2+Jp'q'+Kq'^2)\cdot$$

The terms in p'^2 destroy each other, and the form thus is

$$\phi-\omega_1p' = -V(p'q''-p''q')-\tfrac{1}{V}p'q'(Lp'+Mq'),$$

where

$$L=-\tfrac{J}{G}+\tfrac{2IF}{GE},\qquad M=-\tfrac{K}{G}+\tfrac{I}{E};$$

and, upon substituting herein for I, J, K their values, we find

$$L=-GE_2+EG_2+\tfrac{2F^2E_2}{E}-2FF_2-FE_2+2EF_2-\tfrac{EFG_2}{G},$$

$$M=-GE_2+EG_2-\tfrac{2F^2G_1}{G}+2FF_1+FG_1-2GF_1+\tfrac{GFE_2}{E};$$

viz., these are the values denoted above by the same letters L, M. The final result thus is

$$\frac{1}{\rho}=\frac{q'\sqrt{G}}{R}-\frac{p'\sqrt{E}}{S}=-\phi+\omega_1p',\qquad =\theta-\omega_2q',$$

where the meanings of the symbols have been already explained. A formula substantially equivalent to this, but in a different (and scarcely properly explained) notation, is given, Aoust, “Théorie des coordonnées curvilignes quelconques,” *Annali di Matem.*, t. ii. (1868), pp. 39–64; and I was, in fact, led thereby to the foregoing further investigation.

As to the definition of the radius of geodesic curvature, I remark that, for a curve on a given surface, if PQ be an infinitesimal arc of the curve, then if from Q we let fall the perpendicular QM on the tangent plane at P (the point M being thus a point on the tangent PT of the curve), and if from M, in the tangent plane and at right angles to the tangent, we draw MN to meet the osculating plane of the curve in N, then MN is in fact equal to the infinitesimal arc QQ' mentioned near the beginning of the present paper, and the radius of geodesic curvature ρ is thus a length such that $2\rho\cdot MN=PQ^2$.

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ON THE GAUSSIAN THEORY OF SURFACES.

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In the Memoir, Bour, "Théorie de la déformation des surfaces" (*Jour. de l'Éc. Polyt.*, Cah. 39 (1862), pp. 1–148), the author, working with the form $ds^2 = dv^2 + g^2 du^2$ as a special case of Gauss's formula $ds^2 = E\,dp^2 + 2F\,dp\,dq + G\,dq^2$, obtains (p. 29) the following equations which he calls *fundamental*:—

$$\text{[IV.]}\quad \begin{cases} \dfrac{1}{g}\dfrac{dg_1}{dv} = T^2 - HH_1, \\ \dfrac{dT}{du} + \dfrac{d \cdot Hg}{dv} - H_1 g_1 = 0, \\ \dfrac{d \cdot Tg^2}{dv} + g\dfrac{dH_1}{du} = 0, \end{cases}$$

where g_1 is written to denote $\dfrac{dg}{dv}$, and where (see p. 26)

> H is the curvature of the normal section containing the tangent to the curve $v =$ constant,
>
> H_1 is the curvature of the normal section at right angles to the preceding, containing the tangent to the (geodesic) curve $u =$ constant,
>
> T is the torsion of the same geodesic curve;

or, what is the same thing (see p. 25), the quadric equation for the determination of the principal radii of curvature at the point of the surface is

$$\left(\frac{1}{\rho} - H\right)\left(\frac{1}{\rho} - H_1\right) - T^2 = 0.$$

Writing for greater convenience K in place of the suffixed letter H_1, also V instead of g, so that the differential formula is $ds^2 = dv^2 + V^2du^2$, the equations become

$$\begin{cases} \dfrac{1}{V}\dfrac{d^2V}{dv^2} = T^2 - HK, \\ \dfrac{dT}{du} + \dfrac{d\cdot HV}{dv} - K\dfrac{dV}{dv} = 0, \\ \dfrac{d\cdot TV^2}{dv} + V\dfrac{dK}{du} = 0; \end{cases}$$

or, if we use the suffix 1 to denote differentiation in regard to v, and the suffix 2 to denote differentiation in regard to u, then the equations are

$$\frac{V_{11}}{V} = T^2 - HK,$$

or, what is the same thing,

$$\begin{cases} T_2 + H_1V + (H-K)V_1 = 0, \\ T_1V + 2TV_1 + K_2 = 0. \end{cases}$$

I wish to show how these formulae connect themselves with formulae belonging to the general form $ds^2 = E\,dp^2 + 2F\,dp\,dq + G\,dq^2$. These involve not only Gauss's coefficients E, F, G, but also the coefficients E', F', G' belonging to the inflexional tangents; and, for convenience, I quote the system of definitions, Salmon's *Geometry of Three Dimensions*, 3rd ed., 1874, p. 251, viz.

$$dx,\ dy,\ dz = a\,dp + a'dq,\ b\,dp + b'dq,\ c\,dp + c'dq;$$

$$\begin{aligned} d^2x &= \alpha\,dp^2 + 2\alpha'dp\,dq + \alpha''dq^2, \\ d^2y &= \beta\,dp^2 + 2\beta'dp\,dq + \beta''dq^2, \\ d^2z &= \gamma\,dp^2 + 2\gamma'dp\,dq + \gamma''dq^2; \end{aligned}$$

$$A,\ B,\ C = bc' - b'c,\ ca' - c'a,\ ab' - a'b; \quad V^2 = EG - F^2;$$

$$E' = A\alpha + B\beta + C\gamma, \quad F' = A\alpha' + B\beta' + C\gamma', \quad G' = A\alpha'' + B\beta'' + C\gamma'',$$

so that E', F', G' are, in fact, the determinants

$$\begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ \alpha, & \beta, & \gamma \end{vmatrix}, \quad \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ \alpha', & \beta', & \gamma' \end{vmatrix}, \quad \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix}.$$

The equation for the determination of the principal radii of curvature is

$$(E'\rho - EV)(G'\rho - GV) - (F'\rho - FV)^2 = 0.$$

which, in the particular case $F = 0$ (and therefore $V^2 = EG$), becomes

$$(E'\rho - EV)(G'\rho - GV) - F'^2\rho^2 = 0,$$

or, as this may be written,

$$EGV^2\left(\frac{1}{\rho} - \frac{E'}{EV}\right)\left(\frac{1}{\rho} - \frac{G'}{GV}\right) = F'^2,$$

an equation which corresponds with Bour's form

$$\left(\frac{1}{\rho} - K\right)\left(\frac{1}{\rho} - H\right) - T^2 = 0,$$

and becomes identical with it, if

$$E' = EVK, \quad G' = GVH, \quad F' = -V^2T.$$

But, making p, q correspond to Bour's variables, p to v, and q to u, it is necessary to show that the foregoing values (and not the interchanged values $E' = GVH$, $G' = EVK$) are the correct ones. We have, Salmon, p. 254,

$$\begin{vmatrix} dq, & \rho E' - VE, & \rho F' - VF \\ -dp, & \rho F' - VF, & \rho G' - VG \end{vmatrix} = 0;$$

or, putting herein $F = 0$, the equations may be written

$$\frac{dq}{-dp} = \frac{E'\left(\dfrac{1}{\rho} - \dfrac{VE}{\rho E'}\right)}{F'V} = \frac{F'V}{G'\left(\dfrac{1}{\rho} - \dfrac{VG}{\rho G'}\right)};$$

or, we see that to $dq = 0$ corresponds the value $\dfrac{1}{\rho} = \dfrac{E'}{VE}$, and to $dp = 0$ the value $\dfrac{1}{\rho} = \dfrac{G'}{VG}$. Hence the former of these values $\dfrac{1}{\rho}$ corresponds to $du = 0$, that is, to his $\dfrac{1}{\rho} = K$; and the latter to Bour's $dv = 0$, that is, to his $\dfrac{1}{\rho} = H$; or the values are, as stated,

$$E' = EVK, \quad G' = GVH.$$

The formula $ds^2 = E\,dp^2 + 2F\,dp\,dq + G\,dq^2$ agrees with Bour's $ds^2 = dv^2 + g^2du^2$, if $p = u$, $q = v$, $E = 1$, $F = 0$, $G = g^2$. With these values, $V^2 = EG - F^2 = g^2$, or say $g = V$, and Bour's equation is, as it was before written, $ds^2 = dv^2 + V^2du^2$. And we have to find the three equations which, putting therein $p = u$, $q = v$, $E = 1$, $F = 0$, $G = V^2$, $E' = VK$, $F' = -VT$, $G' = VH$, reduce themselves to Bour's equations.

The first of these is nothing else than the equation for the measure of curvature, viz. Salmon, p. 262 (but, using the suffixes 1 and 2 to denote differentiation in regard to p and q respectively), this is

$$4\,(E'G' - F'^2) = E\,(E_2G_2 - 2F_2G_1 + G_1^2)$$
$$+F\,(E_1G_2 - E_2G_1 - 2E_2F_2 + 4F_1F_2 - 2F_1G_1)$$
$$+G\,(E_1G_1 - 2E_1F_2 + E_2^2)$$
$$-2\,(EG - F^2)(E_{22} - 2F_{12} + G_{11}).$$

In fact, writing herein $E = 1$, $F = 0$, and therefore the differential coefficients of E and F each $= 0$, the equation becomes

$$4\,(E'G' - F'^2) = G_1^2 - 2GG_{11},$$

which is

$$4V^4(HK - T^2) = (2VV_1)^2 - 2V^2(2V_1^2 + 2VV_{11}), \quad = -4V^3V_{11};$$

or finally it is

$$V_{11} = V(T^2 - HK).$$

The other two of Bour's equations are derived from equations which give respectively the values of $E_2' - F_1'$ and $F_2' - G_1'$; viz. starting from the equations

$$E' = A\alpha + B\beta + C\gamma,$$
$$F' = A\alpha' + B\beta' + C\gamma',$$
$$G' = A\alpha'' + B\beta'' + C\gamma'',$$

we see at once that E_2' and F_1' contain, E_2' the terms $A\alpha_2 + B\beta_2 + C\gamma_2$, and F_1' the terms $A\alpha_1' + B\beta_1' + C\gamma_1'$, which are equal to each other (since $\alpha_2 = \alpha_1'$, α and α' are the differential coefficients x_{11}, x_{12} of x, and so β_2 and β_1', γ_2 and γ_1'). Hence

$$E_2' - F_1' = A_2\alpha + B_2\beta + C_2\gamma - A_1\alpha' - B_1\beta' - C_1\gamma'$$

and similarly

$$F_2' - G_1' = A_2\alpha' + B_2\beta' + C_2\gamma' - A_1\alpha'' - B_1\beta'' - C_1\gamma''.$$

Here, from the values of A, B, C, we have

$$\begin{array}{lll}
A = bc' - cb'; & A_1 = \beta c' + b\gamma' - \gamma b' - c\beta'; & A_2 = \beta' c' - \gamma' b' + b\gamma'' - c\beta'' \\
B = ca' - ac'; & B_1 = \gamma a' + c\alpha' - \alpha c' - a\gamma'; & B_2 = \gamma' a' - \alpha' c' + c\alpha'' - a\gamma'' \\
C = ab' - ba'; & C_1 = \alpha b' + a\beta' - \beta a' - b\alpha'; & C_2 = \alpha' b' - \beta' a' + a\beta'' - b\alpha''
\end{array}$$

and, substituting, we find

$$E_2' - F_1' = 2\alpha'\alpha\alpha' + \alpha\alpha''\alpha,$$
$$F_2' - G_1' = -2\alpha\alpha'\alpha'' - \alpha'\alpha''\alpha,$$

if, for shortness, $\alpha'\alpha\alpha'$ denotes the determinant

$$\begin{vmatrix} \alpha', & \alpha, & \alpha \\ b', & \beta, & \beta' \\ c', & \gamma, & \gamma' \end{vmatrix},$$

and so for the other like symbols. Observe that, with

$$\begin{matrix} a, & a', & \alpha, & \alpha', & \alpha'' \\ b, & b', & \beta, & \beta', & \beta'' \\ c, & c', & \gamma, & \gamma', & \gamma'' \end{matrix},$$

we have in all 10 determinants, viz. these are $aa'\alpha, = E'$; $aa'\alpha', = F'$; $aa'\alpha'', = G'$; $\alpha\alpha'\alpha''$; and the six determinants $\alpha\alpha\alpha'$, $\alpha\alpha'\alpha''$, $\alpha\alpha''\alpha$; $\alpha'\alpha\alpha'$, $\alpha'\alpha'\alpha''$, $\alpha'\alpha''\alpha$. The foregoing expressions of $E_2' - F_1'$ and $F_2' - G_1'$ respectively, substituting therein for the determinants $\alpha'\alpha\alpha'$, $\alpha\alpha''\alpha$, $\alpha\alpha'\alpha''$, $\alpha'\alpha''\alpha$ their values as about to be obtained, are the required two equations. We have

$$\begin{aligned}
&aa + bb + cc = E, && aa' + bb' + cc' = F,\\
&a'a + b'b + c'c = F, && a'a' + b'b' + c'c' = G,\\
&\alpha a + \beta b + \gamma c = \tfrac{1}{2}E_1, && \alpha a' + \beta b' + \gamma c' = F_1 - \tfrac{1}{2}E_2,\\
&\alpha' a + \beta' b + \gamma' c = \tfrac{1}{2}E_2, && \alpha' a' + \beta' b' + \gamma' c' = \tfrac{1}{2}G_1,\\
&\alpha'' a + \beta'' b + \gamma'' c = F_2 - \tfrac{1}{2}G_1, && \alpha'' a' + \beta'' b' + \gamma'' c' = \tfrac{1}{2}G_2,
\end{aligned}$$

and if from the first five equations, regarded as equations linear in (a, b, c), we eliminate these quantities, and from the second five equations, regarded as linear in (a', b', c'), we eliminate these quantities, we obtain two sets each of five equations.

$$\begin{vmatrix}
a, & \alpha, & \alpha, & \alpha', & \alpha''\\
b, & b', & \beta, & \beta', & \beta''\\
c, & c', & \gamma, & \gamma', & \gamma'\\
E, & F, & \tfrac{1}{2}E_1, & \tfrac{1}{2}E_2, & F_1 - \tfrac{1}{2}G_1
\end{vmatrix} = 0, \qquad
\begin{vmatrix}
a, & a', & \alpha, & \alpha', & \alpha''\\
b, & b', & \beta', & \beta'', & \\
c, & c', & \gamma, & \gamma', & \gamma''\\
F, & G, & F_1 - \tfrac{1}{2}E_2, & \tfrac{1}{2}G_1, &
\end{vmatrix} = 0.$$

These may be written,

$$F\alpha\alpha'\alpha'' - \tfrac{1}{2}E_1\alpha'\alpha\alpha'' - \tfrac{1}{2}E_2\alpha'\alpha''\alpha - (F_1 - \tfrac{1}{2}G_1)\alpha'\alpha\alpha' = 0,$$

$$-E\alpha\alpha'\alpha'' + \tfrac{1}{2}E_1\alpha\alpha'\alpha'' + \tfrac{1}{2}E_2\alpha\alpha''\alpha + (F_1 - \tfrac{1}{2}G_1)\alpha\alpha\alpha' = 0,$$

$$E\alpha'\alpha'\alpha'' - F\alpha\alpha'\alpha'' + \tfrac{1}{2}E_1G' - (F_1 - \tfrac{1}{2}G_1)F' = 0,$$

$$E\alpha'\alpha''\alpha - F\alpha\alpha''\alpha - \tfrac{1}{2}E_1G' + (F_1 - \tfrac{1}{2}G_1)E' = 0,$$

$$E\alpha'\alpha\alpha - F\alpha\alpha\alpha' + \tfrac{1}{2}E_2F' - \tfrac{1}{2}E_1E' = 0;$$

and

$$G\alpha\alpha'\alpha'' - (F_2 - \tfrac{1}{2}E_2)\alpha'\alpha\alpha'' - \tfrac{1}{2}G_1\alpha'\alpha''\alpha - \tfrac{1}{2}G_2\alpha'\alpha\alpha' = 0,$$

$$-F\alpha\alpha\alpha'' + (F_2 - \tfrac{1}{2}E_2)\alpha\alpha'\alpha'' + \tfrac{1}{2}G_1\alpha\alpha''\alpha + \tfrac{1}{2}G_2\alpha\alpha\alpha' = 0,$$

$$F\alpha'\alpha'\alpha'' - G\alpha\alpha\alpha'' + \tfrac{1}{2}G_1G' - \tfrac{1}{2}G_2F' = 0,$$

$$F\alpha'\alpha''\alpha - G\alpha\alpha'\alpha - (F_2 - \tfrac{1}{2}E_2)G' + \tfrac{1}{2}G_1E' = 0,$$

$$F\alpha'\alpha\alpha - G\alpha\alpha\alpha + (F_2 - \tfrac{1}{2}E_2)F' - \tfrac{1}{2}G_2E' = 0.$$

Attending in each set only to the third, fourth, and fifth equations, and combining these in pairs, we obtain

$$V^2\alpha\alpha'\alpha'' + (\tfrac{1}{2}FG_1 - FF_1 + \tfrac{1}{2}EG_1)F' + (-\tfrac{1}{2}EG_1 + \tfrac{1}{2}FE_2)G' = 0,$$

$$V^2\alpha'\alpha'\alpha'' + (\tfrac{1}{2}GG_1 - GF_1 + \tfrac{1}{2}FG_2)F' + (-\tfrac{1}{2}FG_1 + \tfrac{1}{2}GE_2)G' = 0;$$
$$V^2\alpha\alpha''\alpha + (-\tfrac{1}{2}FE_1 + EF_1 - \tfrac{1}{2}EE_2)G' + (-\tfrac{1}{2}FG_1 + FF_1 - \tfrac{1}{2}EG_1)E' = 0,$$
$$V^2\alpha'\alpha''\alpha + (-\tfrac{1}{2}GE_1 + FF_1 - \tfrac{1}{2}FE_2)G' + (-\tfrac{1}{2}GG_1 + GF_1 - \tfrac{1}{2}FG_2)E' = 0;$$
$$V^2\alpha\alpha\alpha' + (\tfrac{1}{2}EG_2 - \tfrac{1}{2}FE_2)E' + (\tfrac{1}{2}FE_1 - EF_1 + \tfrac{1}{2}EE_2)F' = 0,$$
$$V^2\alpha'\alpha\alpha' + (\tfrac{1}{2}FG_2 - \tfrac{1}{2}GE_2)E' + (\tfrac{1}{2}GE_1 - FF_1 + \tfrac{1}{2}FE_2)F' = 0.$$

We thus obtain

$$E_2' - F_1' = \tfrac{2}{V^2}[(-\tfrac{1}{2}FG_1 + \tfrac{1}{2}GE_2)E' + (-\tfrac{1}{2}GE_1 + FF_1 - \tfrac{1}{2}FF_2)F']$$

$$+\tfrac{1}{V^2}\{(\tfrac{1}{2}FE_1 - EF_1 + \tfrac{1}{2}EE_2)G' + (\tfrac{1}{2}FG_1 - FF_1 + \tfrac{1}{2}EG_1)E'\},$$

$$F_2' - G_1' = \tfrac{2}{V^2}\{(\tfrac{1}{2}FG_1 - FF_1 + \tfrac{1}{2}EG_1)F' + (-\tfrac{1}{2}EG_1 + \tfrac{1}{2}FE_2)G'\}$$

$$+\tfrac{1}{V^2}\{(-\tfrac{1}{2}GE_1 + FF_1 - \tfrac{1}{2}FE_2)G' + (-\tfrac{1}{2}GG_1 + GF_1 - \tfrac{1}{2}FG_2)E'\};$$

or, finally,

$$E_2' - F_1' = \tfrac{1}{V^2}\{(-\tfrac{1}{2}FG_1 + GE_2 - FF_1 + \tfrac{1}{2}EG_2)E'$$

$$+(-GE_1 + 2FF_1 - FF_2)F' + (\tfrac{1}{2}FE_1 - EF_1 + \tfrac{1}{2}EE_2)G'\},$$

$$F_2' - G_1' = \tfrac{1}{V^2}\{(-\tfrac{1}{2}GG_1 + GF_1 - \tfrac{1}{2}FG_2)E'$$

$$+(FG_1 - 2FF_1 + EG_2)F' + (-\tfrac{1}{2}GE_1 + FF_1 - EG_1 + \tfrac{1}{2}FE_2)G'\},$$

which are the required formulae; and which may, I think, be regarded as new formulae in the Gaussian theory of surfaces.

Writing herein as before, the first of these becomes

$$V^2(VK)_2 + (V^2T)_1 = \tfrac{1}{V^2}[\tfrac{1}{2}(V^2)_1VK] = V_2K,$$

that is,

$$V^2V_2K + V^3K_2 + V^2T_1 + 2VV_1T = V_2K;$$

or finally

$$VT_1 + 2TV_1 + K_2 = 0,$$

which is Bour's third equation. And the second equation becomes

$$-(V^2T)_2 - (V^2H)_1 = \tfrac{1}{V^2}\{-\tfrac{1}{2}V^2V_1VK + (V^2)_1(-V^2T)(V^2)_1VH\}$$

that is,

$$= -V^2V_1K - 2VV_1T - 2V^2V_1H,$$

or finally

$$-V^2T_2 - 2VV_2T - V^2H_1 - 3V^2V_1H = -V^2V_1K - 2VV_1T - 2V^2V_1H;$$

$$T_2 + VH_1 + (H-K)V_1 = 0,$$

which is Bour's second equation.

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NOTE ON LANDEN'S THEOREM.

[From the *Proceedings of the London Mathematical Society*, vol. xiii. (1882), pp. 47, 48.
Read November 10, 1881.]

Landen's theorem, as given in the paper "An Investigation of a General Theorem for finding the length of any Arc of any Conic Hyperbola by means of two Elliptic Arcs, with some other new and useful Theorems deduced therefrom," *Phil. Trans.*, t. lxv. (1775), pp. 283–289, is, as appears by the title, a theorem for finding the length of a hyperbolic arc in terms of the length of two elliptic arcs; this theorem being obtained by means of the following differential identity, viz., if

$$t = gx\sqrt{\frac{m^2 - x^2}{m^2 - g^2x^2}}$$

where

$$g = \frac{m^2 - n^2}{n^2},$$

then

$$\sqrt{\frac{m^2 - gx^2}{m^2 - x^2}}\,dx = \left\{\tfrac{1}{2} + \tfrac{1}{4}\sqrt{\frac{(m-n)^2 - t^2}{(m+n)^2 - t^2}} + \tfrac{1}{4}\sqrt{\frac{(m+n)^2 - t^2}{(m-n)^2 - t^2}}\right\} dt,$$

(this is exactly Landen's form, except that he of course writes $\dot{x}$, $\dot{t}$ respectively); viz., integrating each side, and interpreting geometrically in a very ingenious and elegant manner the three integrals which present themselves, he arrives at his theorem for the hyperbolic arc; but with this I am not now concerned.

Writing for greater convenience $m = 1$, $n = k'$, and therefore $g = k^2$, if as usual $k^2 + k'^2 = 1$, the transformation is

$$t = k^2x\sqrt{\frac{1 - x^2}{1 - k^4x^2}},$$

leading to

$$\sqrt{\frac{1-k^2x^2}{1-x^2}}\,dx = \left\{\tfrac{1}{2} + \tfrac{1}{4}\sqrt{\frac{(1-k')^2-t^2}{(1+k')^2-t^2}} + \tfrac{1}{4}\sqrt{\frac{(1+k')^2-t^2}{(1-k')^2-t^2}}\right\} dt.$$

The form in which the transformation is usually employed (see my *Elliptic Functions*, pp. 177, 178) is

$$y = (1+k')x\sqrt{\frac{1-x^2}{1-k^2x^2}},$$

leading to

$$\frac{(1+k')\,dx}{\sqrt{1-x^2\cdot 1-k^2x^2}} = \frac{dy}{\sqrt{1-y^2\cdot 1-\lambda^2y^2}},$$

where

$$\lambda = \frac{1-k'}{1+k'}.$$

If, to identify the two forms, we write $y = \dfrac{t}{1-k'}$, and in the last equation introduce t in place of y, the last equation becomes

$$\frac{dx}{\sqrt{1-x^2\cdot 1-k^2x^2}} = \frac{dt}{\sqrt{\{(1-k')^2-t^2\}\{(1+k')^2-t^2\}}}.$$

Comparing with Landen's form, in order that the two may be identical, we must have

$$1-k^2x^2 = \left\{\tfrac{1}{2} + \tfrac{1}{4}\sqrt{\frac{(1-k')^2-t^2}{(1+k')^2-t^2}} + \tfrac{1}{4}\sqrt{\frac{(1+k')^2-t^2}{(1-k')^2-t^2}}\right\}\times\sqrt{(1-k')^2-t^2}\sqrt{(1+k')^2-t^2},$$

viz., this is

$$1-k^2x^2 = \tfrac{1}{4}\left[\sqrt{(1-k')^2-t^2} + \sqrt{(1+k')^2-t^2}\right]^2,$$

that is,

$$1-k^2x^2 = \tfrac{1}{2}\left[1+k'^2-t^2+\sqrt{\{(1-k')^2-t^2\}\{(1+k')^2-t^2\}}\right],$$

where the function under the radical sign is

$$k'^4 - 2(1+k'^2)t^2 + t^4 \ (= T \text{ suppose})$$

and this must consequently be a form of the original integral equation

$$t = k^2x\sqrt{\frac{1-x^2}{1-k^4x^2}}.$$

In fact, squaring and solving in regard to x^2 with the assumed sign of the radical, we have

$$x^2 = \frac{k'^2+t^2-\sqrt{T}}{2k^2},$$

corresponding to an equation given by Landen. And we thence have

$$1 - k^2x^2 = \tfrac{1}{2}[2 - k'^2 - t^2 + \sqrt{T}], \quad = \tfrac{1}{2}[1 + k^2 - t^2 + \sqrt{T}],$$

which is the required expression for $1 - k^2x^2$.

The trigonometrical form $\sin(2\phi' - \phi) = c\sin\phi$ of the relation between y and x does not occur in Landen; it is employed by Legendre, I believe, in an early paper, *Mém. de l'Acad. de Paris*, 1786, and in the *Exercices*, 1811, and also in the *Traité des Fonctions Elliptiques*, 1825, and by means of it he obtains an expression for the arc of a hyperbola in terms of two elliptic functions, $E(c, \phi)$, $E(c', \phi')$, showing that the arc of the hyperbola is expressible by means of two elliptic arcs, this, he observes, "est le beau théorème dont Landen a enrichi la géométrie." We have, then (1828), Jacobi's proof, by two fixed circles, of the addition-theorem (see my *Elliptic Functions*, p. 28), and the application of this (p. 30) to Landen's theorem is also due to Jacobi, see the "Extrait d'une lettre adressée à M. Hermite," *Crelle*, t. xxxii. (1846), pp. 176–181; the connection of the demonstrations, by regarding the point, which is alone necessary for Landen's theorem as the limit of the smaller circle in the figure for the addition-theorem is due to Durege (see his *Theorie der elliptischen Functionen*, Leipzig, 1861, pp. 168, et seq.).

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ON A FORMULA RELATING TO ELLIPTIC INTEGRALS OF THE THIRD KIND.

[From the *Proceedings of the London Mathematical Society*, vol. xiii. (1882), pp. 175, 176. Presented May 11, 1882.]

The formula for the differentiation of the integral of the third kind

$$\Pi = \int_0^\phi \frac{d\phi}{(1 + n\sin^2\phi)\Delta}$$

in regard to the parameter n, see my *Elliptic Functions*, Nos. 174 *et seq.*, may be presented under a very elegant form, by writing therein

$$\sin^2\phi = x = \operatorname{sn}^2 u, \quad \sin\phi\cos\phi\,\Delta = y = \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u,$$

and thus connecting the formula with the cubic curve

$$y^2 = x(1-x)(1-k^2x).$$

The parameter must, of course, be put under a corresponding form, say $n = -\dfrac{1}{a}$, where $a = \operatorname{sn}^2\theta$, $b = \operatorname{sn}\theta \operatorname{cn}\theta \operatorname{dn}\theta$, and therefore (a, b) are the coordinates of the point corresponding to the argument θ. The steps of the substitution may be effected without difficulty, but it will be convenient to give at once the final result and then verify it directly. The result is

$$\frac{d}{d\theta}\frac{b}{a-x} = \frac{d}{du}\frac{y}{x-a} = k^2(a-x).$$

We, in fact, have

$$\frac{d}{du}x = 2\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u = 2y,$$

and thence

$$y\frac{dx}{du} = 2y^2,$$

that is,

$$y\frac{dx}{du} = 2x\big[1 - (1+k^2)x + k^2x^2\big].$$

Also

$$\begin{aligned}\frac{dy}{du} &= \operatorname{cn}^2 u \operatorname{dn}^2 u - \operatorname{sn}^2 u \operatorname{dn}^2 u - k^2 \operatorname{sn}^2 u \operatorname{cn}^2 u \\ &= 1 - 2(1+k^2)x + 3k^2x^2,\end{aligned}$$

and hence

$$\frac{d}{du}\frac{y}{x-a} = \frac{1}{(a-x)^2}\left\{(a-x)\frac{dy}{du} - y\frac{dx}{du}\right\}$$

$$= \frac{1}{(a-x)^2}\big[-x - a + 2(1+k^2)ax + k^2x^3 - 3k^2ax^2\big].$$

Interchanging the letters, we have

$$\frac{d}{d\theta}\frac{b}{a-x} = \frac{1}{(a-x)^2}\big[-x - a + 2(1+k^2)ax + k^2a^3 - 3k^2a^2x\big],$$

and hence, subtracting,

$$\begin{aligned}\frac{d}{d\theta}\frac{b}{a-x} - \frac{d}{du}\frac{y}{x-a} &= \frac{1}{(a-x)^2}\left\{k^2(a-x)^3\right\} \\ &= k^2(a-x),\end{aligned}$$

which is the required result.

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ON THE 34 CONCOMITANTS OF THE TERNARY CUBIC.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 1–15.]

I have, (by aid of Gundelfinger's formulae, afterwards referred to), calculated, and I give in the present paper, the expressions of the 34 concomitants of the canonical ternary cubic $ax^3 + by^3 + cz^3 + 6lxyz$, or, what is the same thing, the 34 covariants of this cubic and the adjoint linear function $\xi x + \eta y + \zeta z$: this is the chief object of the paper. I prefix a list of memoirs, with short remarks upon some of them; and, after a few observations, proceed to the expressions for the 34 concomitants; and, in conclusion, exhibit the process of calculation of these concomitants other than such of them as are taken to be known forms. I insert a supplemental table of 6 derived forms.

The list of memoirs (not by any means a complete one) is as follows:

Hesse, Ueber die Elimination der Variabeln aus drei algebraischen Gleichungen vom zweiten Grade mit zwei Variabeln: *Crelle*, t. xxviii. (1844), pp. 68–96. Although purporting to relate to a different subject, this is in fact the earliest, and a very important, memoir in regard to the general ternary cubic; and in it is established the canonical form, as Hesse writes it, $y_1^3 + y_2^3 + y_3^3 + 6my_1y_2y_3$.

Aronhold, Zur Theorie der homogenen Functionen dritten Grades von drei Variabeln: *Crelle*, t. xxxix. (1850), pp. 140–159.

Cayley, Third Memoir on Quantics: *Phil. Trans.*, t. cxlvi. (1856), pp. 627–647 [144].

Aronhold, Theorie der homogenen Functionen dritten Grades von drei Variabeln: *Crelle*, t. lv. (1858), pp. 97–191.

Salmon, Lessons Introductory to the Modern Higher Algebra: 8°, Dublin, 1859.

Cayley, Seventh Memoir on Quantics: *Phil. Trans.*, t. cli. (1861), pp. 277–292 [269].

Brioschi, Sur la théorie des formes cubiques à trois indéterminées: *Comptes Rendus*, t. lvi. (1863), pp. 304–307.

Hermite, Extrait d'une lettre à M. Brioschi: *Crelle*, t. lxiii. (1864), pp. 30–32, followed by a note by Brioschi, pp. 32–33.

The skew covariant of the ninth order, which is $y^3-z^3\cdot z^3-x^3\cdot x^3-y^3$ for the canonical form $x^3+y^3+z^3+6lxyz$, and the corresponding contravariant $\eta^3-\zeta^3\cdot\zeta^3-\xi^3\cdot\xi^3-\eta^3$, alluded to p. 116 of Salmon's *Lessons*, were obtained, the covariant by Brioschi and the contravariant by Hermite, in the last-mentioned papers.

Clebsch and Gordan, Ueber die Theorie der ternären cubischen Formen: *Math. Annalen*, t. i. (1869), pp. 56–89.

The establishment of the complete system of the 34 covariants, contravariants and *Zwischenformen*, or, as I have here called them, the 34 concomitants, was first effected by Gordan in the next following memoir:

Gordan, Ueber die ternären Formen dritten Grades: *Math. Annalen*, t. i. (1869), pp. 90–128.

And the theory is farther considered:

Gundelfinger, Zur Theorie der ternären cubischen Formen: *Math. Annalen*, t. vi. (1871), pp. 144–163. The author speaks of the 34 forms as being "theils mit den von Gordan gewählten identisch, theils möglichst einfache Combinationen derselben." They are, in fact, the 34 forms given in the present paper for the canonical form of the cubic, and the meaning of the adopted combinations of Gordan's forms will presently clearly appear.

There is an advantage in using the form $ax^3+by^3+cz^3+6lxyz$ rather than the Hessian form $x^3+y^3+z^3+6lxyz$, employed in my Third and Seventh Memoirs on Quantics: for the form $ax^3+by^3+cz^3+6lxyz$ is what the general cubic

$$(a,b,c,f,g,h,i,j,k,l)(x,y,z)^3$$

becomes by no other change than the reduction to zero of certain of its coefficients; and thus any concomitant of the canonical form consists of terms which are leading terms of the same concomitant of the general form.

The concomitants are functions of the coefficients $(a,b,\ldots,l)$, of (ξ,η,ζ), and of (x,y,z): the dimensions in regard to the three sets respectively may be distinguished as the degree, class, and order; and we have thus to consider the deg-class-order of a concomitant.

Two or more concomitants of the same deg-class-order may be linearly combined together: viz., the linear combination is the sum of the concomitants each multiplied by a mere number. The question thus arises as to the selection of a representative concomitant. As already mentioned, I follow Gundelfinger, viz., my 34 concomitants of the canonical form correspond each to each (with only the difference of a numerical factor of the entire concomitant) to his 34 concomitants of the general form. The principle underlying the selection would, in regard to the general form, have to be explained altogether differently; but this principle exhibits itself in a very remarkable manner in regard to the canonical form $ax^3+by^3+cz^3+6lxyz$.

Each concomitant of the general form is an indecomposable function, not breaking up into rational factors; but this is not of necessity the case in regard to a canonical form: only a concomitant which does break up must be regarded as indecomposable, no factor of such concomitant being rejected, or separated. So far from it, there is, in regard to the canonical form in question, a frequent occurrence of $abc + 8l^3$ or a power thereof, either as a factor of a unique concomitant, or when there are two or more concomitants of the same deg-class-order, then as a factor of a properly selected linear combination of such concomitants: and the principle referred to is, in fact, that of the selection of such combination for the representative concomitant; or (in other words) the representative concomitant is taken so as to contain as a factor the highest power that may be of $abc+8l^3$. As to the signification of this expression $abc + 8l^3$, I call to mind that the discriminant of the form is $abc\,(abc + 8l^3)^3$.

As to numerical factors: my principle has been, and is, to throw out any common numerical divisor of all the terms: thus I write $S = -abcl + l^4$, instead of Aronhold's $S = -4abcl + 4l^4$. There is also the question of nomenclature: I retain that of my Seventh Memoir on Quantics, except that I use single letters H, P, &c., instead of the same letters with U, thus HU, PU, &c.; in particular, I use U, H, P, Q instead of Aronhold's f, Δ, S_f, T_f. It is thus at all events necessary to make some change in Gundelfinger's letters; and there is moreover a laxity in his use of accented letters; his B, B', B'', B''', and so in other cases E, E', E'', &c., are used to denote functions derived in a determinate manner each from the preceding one (by the δ-process explained further on); whereas his L, $\bar{L}$; M, M'; N, N' are functions having to each other an altogether different relation; also three of his functions are not denoted by any letters at all. Under the circumstances, I retain only a few of his letters; use the accent where it denotes the δ-process; and introduce barred letters $\bar{J}$, $\bar{K}$, &c., to denote a different correspondence with the unbarred letters J, K, &c. But I attach also to each concomitant a numerical symbol showing its deg-class-order, thus: 541 (degree = 5, class = 4, order = 1) or 1290, (there is no ambiguity in the two-digit numbers 10, 11, 12 which present themselves in the system of the 34 symbols); and it seems to me very desirable that the significations of these deg-class-order symbols should be considered as permanent and unalterable. Thus, in writing $S = 400 = -abcl + l^4$, I wish the 400 to be regarded as denoting its expressed value $-abcl + l^4$: if the same letter S is to be used in Aronhold's sense to denote $-4abcl + 4l^4$, this would be completely expressed by the new definition $S_1 = 4.400$, the meaning of the symbol 400 being explained by reference to the present memoir, or by the actual quotation $400 = -abcl + l^4$.

I proceed at once to the table: for shortness, I omit, in general, terms which can be derived from an expressed term by mere cyclical interchanges of the letters (a, b, c), (ξ, η, ζ), (x, y, z).

Table of the 34 Covariants of the Canonical Cubic $ax^3+by^3+cz^3+6lxyz$ *and the linear form* $\xi x+\eta y+\zeta z$.

First Part, 10 Forms. Class = Order.

Current No.

1 $S = 400 = -abcl + l^4.$

2 $T = 600 = a^2b^2c^2 - 20abcl^3 - 8l^6.$

3 $A = 011 = \xi x + \eta y + \zeta z.$

4 $\Theta = 222 = x^2[-l^2\xi^2 - 2al\eta\zeta]\ldots$
$+ yz[bc\xi^2 + 2l^2\eta\zeta]\ldots$

5 $\Theta' = 422 = x^2[l(abc+2l^3)\xi^2 + a(abc-4l^3)\eta\zeta]\ldots$
$+ yz[6bcl^2\xi^2 - 2l(abc+2l^3)\eta\zeta]\ldots$

6 $\Theta'' = 622 = x^2[-(abc+2l^3)^2\xi^2 + 2al^2(abc-2l^3)\eta\zeta]\ldots$
$+ yz[36bcl^4\xi^2 + 2(abc+2l^3)^2\eta\zeta]\ldots$

7 $B = 333 = x^3[a^2(c\eta^3 - b\zeta^3)]\ldots$
$+ y^2z[(abc+8l^3)\eta^2\zeta + 2bl^2\xi^3 - 6bcl\xi\eta^2]\ldots$
$+ yz^2[-(abc+8l^3)\eta\zeta^2 - 6bcl\xi^2\eta - 12cl^2\xi^3]\ldots$

8 $B' = 533 = x^3[3a^2l^2(c\eta^3 - b\zeta^3)]\ldots$
$+ y^2z[-l^2(abc+8l^3)\eta\zeta^2 + 4bl(-abc+l^3)\xi^2\zeta - bc(abc+10l^3)\xi\eta^2]\ldots$
$+ yz^2[l^2(abc+8l^3)\eta\zeta^2 - bc(abc+10l^3)\xi^2\eta + 4cl(-abc+l^3)\xi\zeta^2]\ldots$

9 $B'' = 733 = x^3[9a^2l^4(c\eta^3 - b\zeta^3)]\ldots$
$+ y^2z[l(abc+8l^3)(2abc+l^3)\eta\zeta^2$
$+ b(abc+2l^3)(abc-10l^3)\xi^2\zeta + 6bcl^2(-abc+l^3)\xi\eta^2]\ldots$
$+ yz^2[-l(abc+8l^3)(2abc+l^3)\eta\zeta^2$
$- 6bcl^2(-abc+l^3)\xi^2\eta + c(abc+2l^3)(abc-10l^3)\xi\zeta^2]\ldots$

10 $B''' = 933 = x^3[27a^2l^6(c\eta^3 - b\zeta^3)]\ldots$
$+ y^2z[-(abc+8l^3)(abc-l^3)^2\eta^2\zeta + 9bl^3(abc-2l^3)^2\xi^2\zeta$
$- 27bcl^4(abc+2l^3)\xi\eta^2]\ldots$
$+ yz^2[(abc+8l^3)(abc-l^3)^2\eta\zeta^2 - 27bcl^4(abc+2l^3)\xi^2\eta$
$- 9cl^3(abc+2l^3)^2\xi\eta\zeta^2]\ldots$

Second Part, $(4+4=)$ *8 forms. Class* $=0$, *and Order* $=0$.

Class $=0$.

11 $U = 103 = ax^3 + by^3 + cz^3 + 6lxyz.$

12 $H = 303 = l^2(ax^3 + by^3 + cz^3) - (abc + 2l^3)xyz.$

13 $\Psi = 806 = (abc + 8l^3)^2(a^2b^2c^2$
$+ b^2y^6 + c^2z^6 - 10(bcy^3z^3 + caz^3x^3 + abx^3y^3)).$

14 $\Omega = 1209 = (abc + 8l^3)^3[by^3 - cz^3 \cdot cz^3 - ax^3 \cdot ax^3 - by^3].$

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Current No. *Order* $= 0$.

15 $P = 330 = -l(bc\xi^3 + ca\eta^3 + ab\zeta^3) + (-abc + 4l^3)\xi\eta\zeta.$

16 $Q = 530 = (abc - 10l^3)(bc\xi^3 + ca\eta^3 + ab\zeta^3) - 6l^2(5abc + 4l^3)\xi\eta\zeta.$

17 $F = 460 = b^2c^2\xi^6 + c^2a^2\eta^6 + a^2b^2\zeta^6$
$$- 2(abc + 16l^3)(a\eta^3\zeta^3 + b\zeta^3\xi^3 + c\xi^3\eta^3)$$
$$- 24l^2(bc\xi^4 + ca\eta^4 + ab\zeta^4)\xi\eta\zeta - 24l(abc + 2l^3)\xi^2\eta^2\zeta^2.$$

18 $\Pi = 1290 = (abc + 8l^3)^3[c\eta^3 - b\zeta^3 \cdot a\xi^3 - c\eta^3 \cdot b\zeta^3 - a\xi^3].$

Third Part, $(8 + 8 =)$ *16 forms. Class less or greater than Order.*

Class less than Order.

19 $J = 413 = (abc + 8l^3)\{\xi x(by^3 - cz^3) + \eta y(cz^3 - ax^3) + \zeta z(ax^3 - by^3)\}.$

20 $K = 613 = (abc + 8l^3)\{\xi[alx^4 - 2blxy^3 - 2clxz^3 - 3bcy^2z^2]\ldots\}.$

21 $K' = 813 = (abc + 8l^3)\{\xi[(abc + 2l^3)(ax^4 - 2bxy^3 - 2cxz^3) - 18bcl^2y^2z^2]\ldots\}.$

22 $E = 625 = (abc + 8l^3)\{\xi(by^3 - cz^3)[2l^2x^2 + bcyz]\ldots$
$$+ \eta\zeta(by^3 - cz^3)[4alx^2 - 2l^2yz]\ldots\}.$$

23 $E' = 825 = (abc + 8l^3)\{\xi(by^3 - cz^3)[l(abc + 2l^3)x^2 - 3bcl^2yz]\ldots$
$$+ \eta\zeta(by^3 - cz^3)[a(abc + 4l^3)x^2 - l(abc + 2l^3)yz]\ldots\}.$$

24 $E'' = 1025 = (abc + 8l^3)\{\xi(by^3 - cz^3)[(abc + 2l^3)^2x^2 + 18bcl^4yz]\ldots$
$$+ \eta\zeta(by^3 - cz^3)[-12al^3(abc + 2l^3)x^2 + (abc + 2l^3)^2yz]\ldots\}.$$

25 $M = 917 = (abc + 8l^3)^2\{\xi(by^3 - cz^3)[5alx^4 - blxy^3 - clxz^3 - 3bcy^2z^2]\ldots\}.$

26 $M' = 1117 = (abc + 8l^3)^2\{\xi(by^3 - cz^3)[(abc + 2l^3)(5ax^4 - bxy^3 - cxz^3)$
$$+ 18bcl^2y^2z^2]\ldots\}.$$

Order less than Class.

27 $\bar{J} = 841 = (abc + 8l^3)^2[x\xi a(c\eta^3 - b\zeta^3) + y\eta b(a\xi^3 - c\zeta^3) + z\zeta c(b\zeta^3 - a\eta^3)].$

28 $\bar{K} = 541 = (abc + 8l^3)\{x[bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3 - 6ab\eta^2\zeta^2]\ldots\}.$

29 $\bar{K}' = 741 = (abc + 8l^3)\{x[l^2(bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3) + a(abc + 2l^3)\eta^2\zeta^2]\ldots\}.$

30 $\bar{E} = 652 = (abc + 8l^3)\{x^2(c\eta^3 - b\zeta^3)[2al\xi^2 - a^2\eta\zeta]\ldots$
$$+ yz(c\eta^3 - b\zeta^3)[bl^2\xi^2 + 2al\eta\zeta]\ldots\}.$$

31 $\bar{E}' = 852 = (abc + 8l^3)\{x^2(c\eta^3 - b\zeta^3)[a(abc - 4l^3)\xi^2 - 6a^2l\eta\zeta]\ldots$
$$+ yz(c\eta^3 - b\zeta^3)[3bl(abc + 2l^3)\xi^2 - a(abc + l^3)\eta\zeta]\ldots\}.$$

32 $\bar{E}'' = 1052 = (abc + 8l^3)\{x^2(c\eta^3 - b\zeta^3)[-3al^2(abc + 2l^3)\xi^2 + 9a^2l^4\eta\zeta]\ldots$
$$+ yz(c\eta^3 - b\zeta^3)[(abc + 2l^3)^2 - 3al^2(abc + l^3)\eta\zeta]\ldots\}.$$

33 $\bar{M} = 771 = (abc + 8l^3)^2\{x(c\eta^3 - b\zeta^3)[(abc - 8l^3)a^2c\xi\eta^3 - a^2b\xi\zeta^3$

$$- 12al^3\xi^2\eta\zeta - 6a^2bl\eta^2\zeta]\ldots\}.$$

34 $$\bar{M}' = 971 = (abc+8l^3)^2\{x(c\eta^3 - b\zeta^3)[l^2(7abc-8l^3)\xi^4 - 3a^2cl^2\xi\eta^3 + 3a^2bl\xi\zeta^3 - 4al(abc+l^3)\xi^2\eta\zeta + a^2(abc+10l^3)\eta^2\zeta^2]\ldots\}.$$

To this may be joined the following *Supplemental Table* of certain Derived Forms:

Current No.

35 $R = 1200 = 64S^3 - T^2 = -abc(abc + 8l^3)^3.$

36 $C = 703 = -TU + 24SH = (abc + 8l^3)\{(-abc + 4l^3)(ax^3 + by^3 + cz^3) + 18abclxyz\}.$

37 $D = 903 = 8S^2U - 3TH = (abc + 8l^3)\{l(5abc + 4l^3)(ax^3 + by^3 + cz^3) + 3abc(abc - 10l^3)xyz\}.$

38 $Y = 930 = 3TP - 4SQ = (abc + 8l^3)^2\{l(bc\xi^3 + ca\eta^3 + ab\zeta^3) - 3abc\xi\eta\zeta\}.$

39 $Z = 1130 = -48S^2P + TQ = (abc + 8l^3)^2\{(abc + 2l^3)(bc\xi^3 + ca\eta^3 + ab\zeta^3) + 18abcl^2\xi\eta\zeta\}.$

40 $\Phi = 1660 = 12(abc + 8l^3)F - 288STP + 768S^2PQ - 8TQ^2 = (abc + 8l^3)^2\{b^2c^2\xi^6 + c^2a^2\eta^6 + a^2b^2\zeta^6 - 10abc(a\eta^3\zeta^3 + b\zeta^3\xi^3 + c\xi^3\eta^3)\};$

viz. these are derived forms characterized by having a power of $abc + 8l^3$ as a factor: R is the discriminant; C, D, Y, Z occur in Aronhold, and in my Seventh memoir on Quantics [269]: Φ in Clebsch and Gordan's memoir of 1869.

I regard as known forms A, U, H, P, Q, S, T, F, that is, the eight forms 3, 11, 12, 15, 16, 1, 2, 17; the remaining 26 forms are expressed in terms of these by formulae involving notations which will be explained, viz. we have

13 $\Psi = 3\,(bc' + b'c - 2ff', \ldots, gh' + g'h - af' - a'f, \ldots)(X, Y, Z)(X', Y', Z').$

14 $\Omega = -\frac{1}{12}\mathrm{Jac}(U^2, H, \Psi).$

18 $\Pi = -3V[\mathrm{Jac}](P, Q, F).$

4 $\Theta = (bc - f^2, \ldots, gh - af, \ldots)(\xi, \eta, \zeta)^2.$

5 $\Theta' = \frac{1}{2}\delta\Theta.$

6 $\Theta'' = \frac{1}{2}\delta^2\Theta.$

7 $B = -\frac{1}{6}\mathrm{Jac}(U, \Theta, A).$

8 $B' = \frac{1}{8}\delta B.$

9 $B'' = \frac{1}{8}\delta^2 B.$

10 $B''' = \frac{1}{16}\delta^3 B.$

19 $J = -\frac{1}{6}\mathrm{Jac}(U, H, A).$

27 $\bar{J} = \frac{1}{4}[\mathrm{Jac}](P, Q, A).$

20 $K = -\frac{1}{4}\{d_\xi\Theta d_x H + d_\eta\Theta d_y H + d_\zeta\Theta d_z H\} - 3UA.$

21 $K' = -\frac{1}{8}(\delta)K.$

28 $\bar{K} = 3\{d_x\Theta d_\xi P + d_y\Theta d_\eta P + d_z\Theta d_\zeta P\} + QA.$

29 $\bar{K}' = \frac{1}{2}(\delta)\bar{K}.$

22 $E = -\frac{1}{3}\mathrm{Jac}(\Theta, U, A).$

23 $E' = -\frac{1}{4}(\delta)E.$

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$$\begin{aligned}
24\ E'' &= \tfrac{1}{8}(\delta^2)E.\\
30\ \bar{E} &= -\tfrac{1}{3}\mathrm{Jac}(\Theta, U, A).\\
31\ \bar{E}' &= -\tfrac{1}{4}(\delta)\bar{E}.\\
32\ \bar{E}'' &= -\tfrac{1}{8}(\delta^2)\bar{E}.\\
25\ M &= \tfrac{1}{8}\mathrm{Jac}(U, \Psi, A).\\
26\ M' &= -(\delta)M.\\
33\ \bar{M} &= -\tfrac{1}{4}[\mathrm{Jac}](P, F, A).\\
34\ \bar{M}' &= \tfrac{1}{8}(\delta)\bar{M}.
\end{aligned}$$

In explanation of the notations, observe that

$$U = ax^3 + by^3 + cz^3 + 6lxyz,$$

$$H = l^2(ax^3 + by^3 + cz^3) - (abc + 2l^3)xyz.$$

Hence, writing $V = 6H = a'x^3 + b'y^3 + c'z^3 + 6l'xyz$, we have

$$= a', b', c', 6al^2, 6bl^2, 6cl^2, -(abc + 2l^3).$$

And this being so, we write

$$X, Y, Z = ax^2 + 2lyz,\ by^2 + 2lzx,\ cz^2 + 2lxy,$$

$$h = a, b, c, f, g,\ ax, by, cz, lx, ly, lz,$$

for $\frac{1}{3}$ of the first differential coefficients, and $\frac{1}{2}$ of the second differential coefficients of U; and in like manner

$$X', Y', Z' = a'x^2 + 2l'yz,\ b'y^2 + 2l'zx,\ c'z^2 + 2l'xy,$$

$$V' = a', b', c', f', g',\ a'x, b'y, c'z, l'x, l'y, l'z,$$

for $\frac{1}{3}$ of the first differential coefficients, and $\frac{1}{2}$ of the second differential coefficients of $6H$.

Jac is written to denote the Jacobian, viz.

$$\mathrm{Jac}(U, H, \Theta) = \begin{vmatrix} d_xU, & d_yU, & d_zU \\ d_xH, & d_yH, & d_zH \\ d_x\Theta, & d_y\Theta, & d_z\Theta \end{vmatrix}$$

and in like manner [Jac] to denote the Jacobian, when the differentiations are in regard to (ξ, η, ζ) instead of (x, y, z): δ is the symbol of the δ-process, or substitution of the coefficients (a', b', c', l') in place of (a, b, c, l); in fact,

$$\delta = a'd_a + b'd_b + c'd_c + l'd_l.$$

δ, δ^2, &c., each operate directly on a function of (a, b, c, l), the (a', b', c', l') of the symbol δ being in the first instance regarded as constants, and being replaced ultimately by their values; for instance,

$$\delta abc = a'bc + ab'c + abc', \quad \delta^2 abc = 2(ab'c' + a'bc' + a'b'c), \quad \delta^3 abc = 6a'b'c'.$$

In several of the formulae, instead of δ or δ^2, the symbol used is (δ) or (δ^2): in these cases, the function operated upon contains the factor $(abc+8l^3)$ or $(abc+8l^3)^2$, and is of the form $(abc+8l^3)(aU+bV+cW)$ or $(abc+8l^3)^2(a^2U+abV+\&c.)$: the meaning is, that the δ or δ^2 is supposed to operate through the $(abc+8l^3)a$, or $(abc+8l^3)^2a^2$, &c., as if this were a constant, upon the U, V, &c., only; thus: $(\delta)\cdot(abc+8l^3)(aU+bV+cW)$ is used to denote $(abc+8l^3)(a\delta U+b\delta V+c\delta W)$. As to this, observe that, operating with δ instead of (δ), there would be the additional terms $U\delta(abc+8l^3)a+\&c.$; we have in this case

$$\delta(abc+8l^3)a = a+(2a'bc+ab'c+abc'+24l^2l')\delta l^3a,$$

$$= 24a^2bcl^2-24al^2(abc+2l^3) = -48al^5;$$

or the rejected terms in fact vanish. For $(\delta^2)\cdot(abc+8l^3)(aU+bV+cW)$, operating with δ^2, we should have, in like manner, terms $U\delta^2(abc+8l^3)a+\&c.$; here

$$\delta^2(abc+8l^3)a = a'^2bc+2aba'c'+2aca'b'+a^2b'c'+24l^4a'l'+24al'^2,$$

which is found to be

$$= 24a(abc+8l^3)(-abcl+l^4), \quad \text{that is,} \ = 24S(abc+8l^3)a;$$

and the terms in question are thus $= 24S(abc+8l^3)(aU+bV+cW)$, viz.

$$(abc+8l^3)(aU+bV+cW)$$

being a covariant, this is also a covariant; that is, in using (δ^2) instead of δ^2, we in fact reject certain covariant terms; or say, for instance, E being a covariant, then $(\delta^2)E$ is also a covariant, but a different covariant. The calculation with (δ) or (δ^2) is more simple than it would have been with δ or δ^2. See post, the calculations of K, K', &c.

I give for each of the 26 covariants a calculation showing how at least a single term of the final result is arrived at, and, in the several cases for which there is a power of $abc+8l^3$ as a factor, showing how this factor presents itself.

Calculations for the 26 Covariants.

13. $\Psi = 3(bc'+b'c+gh'+g'h-2ff',\ldots,g'h+gh'-af'-a'f,\ldots)(X,Y,Z)(X',Y',Z')$

$$= 3((bc'+b'c)yz-2ll'x^2,\ldots,2ll'yz-(af'+a'f)x^2,\ldots)(ax^2+2lyz,\ldots)(a'x^2+2l'yz,\ldots)$$

$$+2^3(a^2+\ldots).$$

The whole coefficient of x^6 is

$$-3ll'aa' + Ta^2, \quad = -3al^2(abc + 2l^3) + Ta^2,$$

viz. the coefficient of a^2 is

$$\begin{aligned} -3l^2(abc + 2l^3) + a^2b^2c^2 - 20abcl^3 - 8l^6 \\ = a^2b^2c^2 + 18abcl^3 + 64l^6 \\ = (abc + 8l^3)^2. \end{aligned}$$

14. $\Omega = \frac{1}{12}\mathrm{Jac}\Big(U, H, \frac{V^2}{4}\Big) = \frac{1}{8}\begin{vmatrix} X, & X', & \frac{1}{2}d_x\Psi \\ Y, & Y', & \frac{1}{2}d_y\Psi \\ Z, & Z', & \frac{1}{2}d_z\Psi \end{vmatrix}$

Here

$$\begin{aligned}YZ' - Y'Z &= (by^2 + 2lzx)(c'z^2 + 2l'xy) + (cz^2 + 2lxy)(b'z^2 + 2l'xy)\\ &\quad -y^2z^2(bc' - b'c) + x^4(2bl' - 2b'l)x^2y^3 - 2(cl' - c'l)\\ &= -2(abc + 8l^3)x(by^3 - cz^3)\\ &= -\tfrac{1}{2}\cdot \partial_x \tfrac{1}{2}(a^2x^4 + 5abxy^3 + 5acx^4z^3).\end{aligned}$$

Hence the whole is

$$\begin{aligned}6y &= -(abc + 8l^3)\{a^2x^4(by^3 - cz^3) + b^2y^4(cz^3 - ax^3) + c^2z^4(ax^3 - by^3)\},\\ &= (abc + 8l^3)(by^3 - cz^3)(cz^3 - ax^3)(ax^3 - by^3).\end{aligned}$$

18. $\Pi = -3V[\mathrm{Jac}](P, Q, F) = -3V \begin{vmatrix} \partial_\xi P, & \partial_\eta P, & \partial_\zeta P\\ \partial_\xi Q, & \partial_\eta Q, & \partial_\zeta Q\\ \partial_\xi F, & \partial_\eta F, & \partial_\zeta F\end{vmatrix}$

viz. if, in this calculation, we write

$6P = a\xi^3 + b\eta^3 + c\zeta^3 + 6l\xi\eta\zeta$, i.e. $a, b, c, l = -6lbc, -6lca, -6lab, -abc + 4l^3$,
$Q = a'\xi^3 + b'\eta^3 + c'\zeta^3 + 6l'\xi\eta\zeta$, $a', b', c', l' = (abc - 10l^3)(bc, ca, ab)$, $l'^3(5abc + 4l^3)$,

then

$$\Pi = -\tfrac{1}{2}\begin{vmatrix} a\xi^2 + 2l\eta\zeta, & \ldots, & a'\xi^2 + 2l'\eta\zeta, & \partial_\xi F\\ b, \ldots = 2l\eta\zeta, & & b\ldots F & \\ \ldots + 2l\eta & & \ldots + 2l'F & \end{vmatrix}.$$

Here

$$\begin{aligned}&(b\eta^2 + 2l\eta\zeta)(c'\zeta^2 + 2l'\eta\zeta) - (b'\eta + 2l'\zeta\xi)(c\zeta^2 + 2l\xi\eta)\\ &= -(bc' - b'c)\eta^2\zeta^2 - 2(bl' - b'l)\eta^3\xi - 2(cl' - c'l)\xi\zeta^3,\end{aligned}$$

or since

$$\begin{aligned}bc' - b'c &= 0,\\ bl' - b'l &= -6lca\cdot l^3(5abc + 4l^3) - (abc - 10l^3)ca(-abc + 4l^3)\\ &= ca\{6l^3(5abc + 4l^3)(abc - 4l^3)(abc - 10l^3)\}\\ &= ca(abc + 8l^3)^2,\end{aligned}$$

and the like for $cl' - c'l$, the expression is

$$= 2(abc + 8l^3)^2(ca\eta^3 - ab\zeta^3)\xi$$

and the whole is thus

$$\begin{aligned}\partial_\xi P + \partial_\eta P + \partial_\zeta P &= -\tfrac{1}{2}(abc + 8l^3)^2\{(ca\eta^3 - ab\zeta^3)\xi + \ldots\}\\ = -\tfrac{1}{2}(abc + 8l^3)^2&\{(ca\eta^3 - ab\zeta^3)[b^2c^2\xi^5 - (abc + \ldots)(b\zeta^3 + c\eta^3)\&c.]\\ &+(ab\zeta^3 - bc\xi^3)[c^2a^2\eta^5 - (abc + 16l^3)(c\xi^3 + a\zeta^3) + \&c.]\\ &+(bc\xi^3 - ca\eta^3)[a^2b^2\zeta^5 + (abc + 16l^3)(a\eta^3\zeta + b\xi^3\zeta) + \&c.]\}.\end{aligned}$$

Here the coefficient of $\xi^5\eta^3$, inside the { }, is

$$ab^2c^3 + bc^2(abc + 16l^3), \quad = 2bc^2(abc + 8l^3),$$

and consequently the whole is

$$= -(abc + 8l^3)^3(bc^2\xi^5\eta^3 - \ldots),$$

$$= -J = -(abc + 8l^3)^3\{(c\eta^3 - b\zeta^3)(a\xi^3 - c\eta^3)(b\zeta^3 - a\eta^3)\}.$$

4. $\Theta = (bc - f^2, \ldots, gh - af, \ldots)(\xi, \eta, \zeta)^2$

$$= (bcyz - l^2x^2)\xi^2 + \ldots 2(l^2yz - alx^2)\eta\zeta + \ldots$$

which are the terms of the final result

$$\Theta = x^2[-l^2\xi^2 - 2al\eta\zeta] + yz[bc\xi^2 + 2l^2\eta\zeta].$$

5 and 6. The δ-process applied to the terms of Θ just written down gives

$$\Theta' = \tfrac{1}{2}\delta\Theta = x^2[-ll'\xi^2 - (al' + a'l)\eta\zeta] + yz[\tfrac{1}{2}(bc' + b'c)\xi^2 + ll'\eta\zeta],$$

$$\Theta'' = \tfrac{1}{2}\delta^2\Theta = x^2[-l'^2\xi^2 - 2a'l'\eta\zeta] + yz[b'c'\xi^2 + 2l'^2\eta\zeta];$$

substituting for a', b', c', l' their values, we have the corresponding terms of Θ' and Θ'' respectively.

7. $B = -\frac{1}{6}\mathrm{Jac}(U, \Theta, A)$, $\begin{vmatrix} Y, & d_y\Theta, & \eta \\ Z, & d_z\Theta, & \zeta \end{vmatrix}$

A term is $\frac{1}{6}(\eta d_z\Theta - \zeta d_y\Theta)$, and if, in this calculation, we write

$$\Theta = (A, B, C, F, G, H)(x, y, z)^2, \quad \text{i.e. } A = -l^2\xi^2 - 2al\eta\zeta, \ldots, F = \tfrac{1}{2}bc\xi^2 + l^2\eta\zeta,$$

then the term is

$$\tfrac{1}{2}\Theta = (ax^2 + 2lyz)\{x \cdot 2(G\eta - H\zeta) + y \cdot 2(A\zeta - B\eta) + z \cdot 2(C\eta - F\zeta)\}.$$

Here

$$Y_{12} = 2(G\eta - H\zeta) = V\{ca\eta^2 + \ldots \zeta(bc\zeta^2 + l^2\eta\zeta)\}, \quad = a(c\eta^3 - b\zeta^3),$$

and hence the whole term in x^3 is $= a^2(c\eta^3 - b\zeta^3)$.

8, 9, 10. The coefficient of $a^2\eta^3$ in B is a^2c, and hence in δB, $\delta^2 B$, $\delta^3 B$ the coefficients of this term are $2a'ac + a^2c'$, $2a'^2c + 4aa'c'$, $6a^2c'$, whence in

$$B', B'', B''' = \tfrac{1}{8}\delta B,\ \tfrac{1}{8}\delta^2 B,\ \tfrac{1}{16}\delta^3 B \text{ respectively,}$$

the coefficients are

$$\tfrac{1}{8}(a^2c' + 2aa'c),\ \tfrac{1}{8}(a^2c + 2aa'c'),\ \tfrac{6}{16}a^2c',$$

$$= 3l^2a^2c,\ 9l^4a^2c,\ 27l^6a^2c \text{ respectively.}$$

19. $J = -\frac{1}{6}\mathrm{Jac}(U, H, A) = -\frac{1}{6}\begin{vmatrix} X, & X', & \xi \\ Y, & Y', & \eta \\ Z, & Z', & \zeta \end{vmatrix}$

A term is $-\frac{1}{6}(YZ' - Y'Z)\xi$, where, as in a previous calculation,

$$YZ' - Y'Z = -2(abc + 8l^3)x(by^3 - cz^3).$$